ELEC473 Assignment 4 - Synthesising the NIOS II Processor

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Abstract

Chapter 1

Introduction

We focus on expanding familiarity of FPGA *FieldProgrammableGateway* engineering through this assignments focus on QSYS [?] and the NIOS-II [?] processor. The NIOS-II being Altera's [?] own 32-bit embedded processor mirco-architecture specifically designed for syntehesis on FPGA, The NIOS-II is a successor of Altera's first configurable 16-bit embedded processor Nios, introduced in 2000. During this assignment we utilize the Cyclone-II DE2 FPGA which we syntehsis designs, bench test and design specifically for during the course of our work.

The NIOS-II hosts the following design parameters: - 32 general-purpose 32-bit registers - Full 32-bit Instruction set, data path, and address space. - Singla-instruction 32 x 32 mulitply and divide producing a 32-bit result.

This soft-core *Asoftmicroprocessorisamicroprocessorcorethatcanbewhollyimplementedusinglo* *endandcommodityvalidations* [?] nature of NIOS-II enables hardware engineers to specify and generate custom NIOS II core, tailed for their own cusom application requirements.

Qsys (now known as platform designer) is a system integration tool for Intel Quartus Prime. It helps connect multiple intellectual property (IP) blocks, processors (like Nios II), and memory interfaces using standard interconnects like Avalon and AXI. It can be utilized for generated HDL.

This assignment sees us in Part A sythesise a NIOS II processor with 20KB of on-chip memory, a timer, a JTAG UART, 8 parallel I/O pins and system ID component. part a sees us interface the 512 KB SRAM and 8 MB SDRAM on the Altera DE2 Board to this design and test it using memory test programs within NIOS II IDE we will show Name and ID numbers in terminal each time the memory is testing.

part b continues on part a requiring us to develop a custom instruction to count the number of leading 1s in the 32 bit number passed into the instruction. and write a custom program to the this custom instruction. I will also develop a test routine in C or assembler utilizing compiler that has the same instruction fucntionality implemented to compare the speed of the custom instruction against software implementation.

Part C we shall utilize PWM to control the inensite of of LEDR10.

Chapter 2

Part A - SRAM & SDRAM

system configuration:

TODO: ADD Block diagram here.

Would flash memory test program over JTAG to the opposited memory location of which tested, ect.. Memory test added to sdram if testing sram and vice versa.

Memory Testing locations:

Region — Origin — Length — End address ———————————————— SRAM — 0x0000000 — 0x80000 (512 KB) — 0x007FFFF sdram — 0x1000020 — 0x7FFFE0 (8MB) — 0x17FFFFF

SRAM Test

```
             <---->   Nios II Memory Test.   <---->
Name: Ryan McKee   ID: 201954749
This software example tests the memory in your system to assure it
is working properly.  This test is destructive to the contents of
the memory it tests. Assure the memory being tested does not contain
the executable or data sections of this code or the exception address
of the system.


----------------------------------
      Memory Test Main Menu
----------------------------------
     a:  Test RAM
     b:  Test Flash
     q:  Exit
----------------------------------
```

```
Select Choice (a-b): [Followed by <enter>]
 -ERROR:
 is an invalid entry.  Please try again

Press enter to continue...


----------------------------------
      Memory Test Main Menu
----------------------------------
     a:  Test RAM
     b:  Test Flash
     q:  Exit
----------------------------------

Select Choice (a-b): [Followed by <enter>]
A
Base address to start memory test: (i.e. 0x800000)
>
0x0000000
End Address:
>
0x007FFFF
```

SDRAM Test Screen Dumps. To test SDRAM, I had to edit linker settings to move all .test, rodata rwdata bus head stack to sram to keep sdram totally free for destructive testing otherwise the test would hang or fail.

so each of the .text and such in settings.bsp of the bsp package needed to be changed to utilize sram instead of sram for proper testing.

```
             <---->   Nios II Memory Test.   <---->
Name: Ryan McKee
Student ID: 201954749
This software example tests the memory in your system to assure it
is working properly.  This test is destructive to the contents of
the memory it tests. Assure the memory being tested does not contain
the executable or data sections of this code or the exception address
of the system.
```

```
----------------------------------
      Memory Test Main Menu
----------------------------------
     a:  Test RAM
     b:  Test Flash
     q:  Exit
----------------------------------

Select Choice (a-b): [Followed by <enter>]
a
Base address to start memory test: (i.e. 0x800000)
>
0x1000000
End Address:
>
0x17FFFFF

Testing RAM from 0x1000000 to 0x17FFFFF
 -Data bus test passed
 -Address bus test passed
 -Byte and half-word access test passed
 -Testing each bit in memory device. . .  passed
Memory at 0x1000000 Okay

Press enter to continue...
```

Chapter 3

Part B - Custom Instruction

Chapter 4

Part C - PWM Module

Bibliography

[1] Wikipedia contributors. Nios ii. https://en.wikipedia.org/wiki/Nios$_I I, 2026. Accessed : 2026 - 07 - 31.$

[2] Wikipedia contributors. Soft microprocessor. https://en.wikipedia.org/wiki/Soft$_m icroprocessor, 2026. Accessed : 2026 - 07 - 31.$